

A Self-Closing Gate Conundrum

#1 Victory Royale

Element C: Documentation and Analysis of Prior Design Attempts

Student G's family needs a self-closing mechanism that closes their family's gate on its own because his family does not want unwanted animals and people to get inside their fence gate because it is a safety concern that could cause injury. This happens because student G's gate can not close on its own.

Criteria

1. Closes a 20 lb gate entirely (High)
2. Durable - Lasts around 5 years (Medium)
3. Lightweight - Less than 15 lbs (Low)
4. Cannot slam gate shut (Low)

Constraints

1. Weather resistant - Rain, snow, hail, lightning, wind, rust, and erosion (High)
2. Affordable - Less than \$150 (High)
3. Measurements - Less than 4 feet tall and 3 feet long (Medium)
4. Can be cleaned with standard cleaning supplies (Low)

Our solution will need to close a 20 lb gate entirely, the current problem right now is that the gate has to be closed by hand. Instead our solution will push the gate back without having the person closing it by hand. It will be durable meaning if someone runs into it by accident it will

not open on its own, and not fall off and break, making it a reliable design. We want our solution to be lightweight so it does not take long to close, because of how heavy it will be, meaning that when it is light it will close quickly. There is one thing that we do not want to happen is to slam the gate closed because there is a little bar that latches on to the gate to keep it from opening on its own and if it slams into it the little bar could bend and break. Our solution will be able to withstand rain, snow, hail, lightning, and wind. We just do not want our solution to rust. We want our solution to be affordable around \$80.

The design requirements will lead to our solution because it matched our stakeholder's criteria. Our Mrs. G agreed that our design requirements were met in our criteria because it was custom to our stakeholder. Our stakeholder informed us that they do not want to spend more than \$150, and they expect it will last about five to seven years. We did not need to add or remove anything to our list.